

**Phishing Website Detection using Machine Learning Techniques
Critical Evaluation and Reproduction Study**

1. Summary of the Source

The Problem Being Addressed

Phishing websites are fraudulent websites that imitate legitimate websites in order to steal sensitive user information, such as usernames, passwords, credit card numbers, and other personal data. As phishing attacks continue to increase, they have become one of the most common cybersecurity threats affecting both individuals and organizations.

Why the Problem is Important

Detecting phishing websites is essential for protecting users from financial loss, identity theft, and unauthorized access to online accounts. Traditional blacklist-based approaches are often unable to detect newly created phishing websites. Therefore, machine learning techniques provide an effective solution by learning patterns that distinguish phishing websites from legitimate ones.

The Proposed Solution

The selected project proposes the use of several supervised machine learning algorithms to classify websites as either phishing or legitimate. The project extracts meaningful cybersecurity features from website URLs and domain

information and uses these features to train multiple classification models. The performance of the models is then compared using several evaluation metrics.

Dataset Used

The project uses a phishing website dataset containing 10,000 website samples. Each sample is represented by 17 engineered features describing different characteristics of the website, such as the presence of an IP address, URL length, URL depth, DNS records, domain age, TinyURL usage, and other security-related attributes. The target variable (Label) indicates whether the website is phishing or legitimate.

Model and Methodology

The methodology consists of several stages, including data inspection, exploratory data analysis (EDA), feature engineering, model training, and performance evaluation. Five machine learning models were evaluated: Decision Tree, Random Forest, Multilayer Perceptron (MLP), XGBoost, and Support Vector Machine (SVM). Their performance was compared using Accuracy, Precision, Recall, F1-Score, ROC-AUC, Matthews Correlation Coefficient (MCC), and Confusion Matrix analysis.

2. Critical Evaluation

Main Claims Made by the Author

The author claims that machine learning techniques can effectively distinguish phishing websites from legitimate websites by using engineered features extracted from URLs and domain characteristics. The project also claims that comparing multiple machine learning models helps identify the most suitable model for phishing website detection.

Are the Claims Supported?

Overall, the experimental results support the author's main claims. All evaluated models achieved relatively high classification performance, demonstrating that the selected features contain useful information for identifying phishing websites.

However, during our reproduction study, we observed a slight difference in the ranking of the best-performing models. While the original notebook reported XGBoost as the model with the highest test accuracy, our implementation achieved slightly better results with the Multilayer Perceptron (MLP). The difference between the two models was very small (approximately 0.2%), indicating that both models perform similarly on this dataset.

Evaluation Methodology

The evaluation methodology is appropriate for the problem. The project uses a train-test split and compares several machine learning algorithms using common classification metrics. In our implementation, the evaluation was extended by including additional performance measures such as Precision, Recall, F1-Score, ROC-AUC, Matthews Correlation Coefficient (MCC), and Confusion Matrix analysis. These metrics provide a more comprehensive assessment of model performance, especially in cybersecurity applications where false positives and false negatives have different practical consequences.

Weaknesses and Limitations

One limitation of the project is that the dataset contains only engineered numerical features rather than raw website content. As a result, the models may not generalize well to phishing techniques that rely on visual imitation or page content.

Another limitation is that the dataset represents a specific collection of phishing websites and may not fully capture newly emerging phishing attacks. Furthermore, the original project provides limited discussion regarding feature importance, model explainability, and robustness against evolving threats.

Are the Conclusions Justified?

The overall conclusions are justified. The experiments demonstrate that machine learning models can effectively detect phishing websites using engineered cybersecurity features. Although our reproduction produced slightly different model rankings, the overall findings remain consistent with the author's conclusions, confirming that machine learning is an effective approach for phishing website detection.

3 Exploratory Data Analysis (EDA)

Before training the machine learning models, an exploratory data analysis (EDA) was performed to better understand the structure and quality of the dataset. The analysis included examining the dataset dimensions, feature types, missing values, duplicate records, class distribution, outliers, and relationships between features.

The dataset contained 10,000 samples with 17 engineered cybersecurity features and one target variable. No missing values were found, indicating that the dataset was complete and did not require imputation. Duplicate records were identified and analyzed to understand their potential impact on the dataset.

The class distribution analysis showed that the dataset was perfectly balanced, with 50% phishing websites and 50% legitimate websites. This eliminated the need for class balancing techniques such as oversampling or undersampling.

Correlation analysis was performed using a heatmap to identify relationships between features and detect potential redundancy. The analysis revealed moderate correlations between some variables but no severe multicollinearity that required removing features.

Boxplots were used to inspect the distribution of numerical features and identify potential outliers. Since the selected machine learning models are generally robust to moderate outliers, no additional preprocessing was required.

Overall, the exploratory data analysis confirmed that the dataset was well prepared for machine learning and provided valuable insights that guided the subsequent feature engineering and model training stages.

4. Feature Engineering Analysis

Feature Engineering

Feature engineering had already been performed in the original project before the machine learning models were trained. Instead of using the raw URL text, the dataset contains engineered numerical features that describe important characteristics of websites. These features include the presence of an IP address, URL length, URL depth, TinyURL usage, domain age, DNS records, iFrame usage, mouse-over behavior, and several other cybersecurity-related attributes.

Overall Assessment

The feature engineering process represents one of the main strengths of the original project. Instead of relying on raw website URLs, the authors transformed domain characteristics into meaningful numerical features that

are directly relevant to phishing detection. This approach simplifies the learning task while preserving important cybersecurity information, enabling the evaluated machine learning models to achieve strong classification performance.

Importance of the Engineered Features

The engineered features were selected because they represent common characteristics of phishing websites. For example, phishing URLs often contain IP addresses instead of domain names, unusually long URLs, multiple nested directories, shortened URLs (TinyURL), and recently registered domains. Features such as DNS records, iFrame usage, and mouse-over behavior capture suspicious website behavior that is frequently associated with phishing attacks. Therefore, these features provide meaningful information that helps machine learning models distinguish between legitimate and phishing websites

Features Used

The following features were used in this project:

- Have_IP
- Have_At
- URL_Length
- URL_Depth
- Redirection
- https_Domain
- TinyURL
- Prefix/Suffix
- DNS_Record
- Web_Traffic
- Domain_Age
- Domain_End

- iFrame
- Mouse_Over
- Right_Click
- Web_Forwards

The target variable (Label) indicates whether a website is phishing or legitimate.

Feature Relevance

Not all features contribute equally to the classification task. During the exploratory data analysis, correlation analysis showed that some features, such as **URL_Length** and **Prefix/Suffix**, had a stronger relationship with the target label than others. However, no feature was removed because each one may contribute complementary information that improves the overall classification performance when combined with other features.

Encoding

No additional encoding was required. All input features were already represented as numerical values in the dataset, making them directly suitable for machine learning algorithms.

Why No Encoding Was Needed

No additional encoding was required because all categorical information had already been converted into numerical values in the original dataset. Binary features were represented using values such as 0 and 1, allowing all machine learning algorithms to process the data directly without further transformation.

Feature Scaling

Feature scaling was not applied. Most features are binary or small integer values, and the original implementation did not include normalization or standardization. The models were able to achieve high classification performance without additional scaling.

Feature Creation

Feature creation had already been completed in the original project. Instead of using raw URLs, the authors extracted meaningful cybersecurity features that capture properties commonly associated with phishing websites. These engineered features simplify the learning process and improve the ability of machine learning models to distinguish phishing websites from legitimate ones.

Feature Selection

The Domain column was excluded from the training process because it contains the original website address as text and cannot be directly used by the selected machine learning algorithms. The remaining engineered features were retained because they provide relevant information for phishing detection.

Dimensionality Reduction

Dimensionality reduction techniques such as Principal Component Analysis (PCA) were not applied. The dataset contains a relatively small number of meaningful and interpretable features, making dimensionality reduction unnecessary.

Feature Redundancy

The correlation analysis performed during the exploratory data analysis showed that some features are moderately correlated. However, no severe redundancy was observed that required removing additional features. If stronger redundancy were detected, techniques such as correlation analysis, variance analysis, or recursive feature elimination could be used to identify and remove redundant features.

Meaningfulness of the Feature Engineering Process

The feature engineering process is meaningful from both mathematical and cybersecurity perspectives. Mathematically, the engineered features provide informative variables that improve the ability of machine learning algorithms to separate phishing websites from legitimate ones. From a cybersecurity perspective, these features represent common characteristics used by attackers, such as suspicious URLs, domain properties, and website behavior, making them highly relevant for phishing detection.

Additional Features

The model could potentially be improved by including additional features such as SSL certificate validity, webpage content analysis, JavaScript behavior, HTML structure, domain registration information, and external reputation services. These additional features may improve the detection of modern phishing websites and increase the model's robustness against evolving attacks.

5. Reproducibility Analysis

The original project was successfully reproduced using the code and dataset provided in the original GitHub repository. All required files were available, including the dataset and the Jupyter Notebook, allowing the experiments to be executed without missing resources.

The notebook executed successfully after installing the required Python libraries. During the reproduction process, only minor adjustments were required, such as adding additional evaluation metrics (Precision, Recall, F1-Score, ROC-AUC, Matthews Correlation Coefficient, and Confusion Matrix) in order to satisfy the requirements of this assignment. These additions did not modify the original machine learning models but provided a more comprehensive evaluation of their performance.

No hidden preprocessing steps were identified. The preprocessing procedure described in the notebook was consistent with the implementation. The dataset was already cleaned and engineered, allowing the models to be trained directly after separating the features from the target variable.

Although the original notebook reported XGBoost as the best-performing model, our reproduction achieved slightly higher accuracy with the Multilayer Perceptron (MLP). The difference between the two models was very small (approximately 0.2%) and is likely the result of differences in software versions, library implementations, or random initialization during training. Despite this minor difference, the overall conclusions of the original project were successfully reproduced.

Overall, the project demonstrated a high level of reproducibility. The provided code, dataset, and documentation were sufficient to reproduce the experiments and obtain results that were highly consistent with those reported in the original work.

6. Experimental Results

Experiments Performed

The experiments were conducted using the phishing website dataset provided in the original GitHub repository. The dataset was first inspected to verify its structure, feature types, missing values, duplicate records, and class distribution. Exploratory Data Analysis (EDA) was then performed to better understand the relationships between the features and the target variable.

Several supervised machine learning models were trained and evaluated using the same training and testing datasets. The evaluated models were:

- Decision Tree
- Random Forest
- Multilayer Perceptron (MLP)
- XGBoost
- Support Vector Machine (SVM)

Modifications Introduced

The original notebook was extended to satisfy the project requirements. Additional evaluation metrics were added, including Precision, Recall, F1-Score, ROC-AUC, Matthews Correlation Coefficient (MCC), and Confusion Matrix analysis. Furthermore, model comparison tables and visualizations were created to provide a clearer comparison between the different machine learning models.

Evaluation Metrics

The following evaluation metrics were used:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

- Matthews Correlation Coefficient (MCC)
- Confusion Matrix

These metrics provide a comprehensive evaluation of classification performance and are particularly important in cybersecurity because they help measure the trade-off between correctly detecting phishing websites and minimizing false alarms.

Results

The experimental results showed that all evaluated machine learning models achieved relatively high classification performance.

Among the evaluated models, the Multilayer Perceptron (MLP) achieved the highest test accuracy in our implementation, followed closely by XGBoost. Random Forest and Decision Tree also performed well, while Support Vector Machine (SVM) achieved slightly lower performance compared to the other models.

Although the original notebook reported XGBoost as the best-performing model, our reproduction study showed that MLP achieved slightly better performance. The difference between the two models was very small and does not change the overall conclusion that both models are highly effective for phishing website detection.

The confusion matrices further demonstrated that the strongest models successfully classified most phishing and legitimate websites while maintaining relatively low classification errors.

7. Conclusions

Key Findings

This project successfully reproduced a machine learning solution for phishing website detection and provided a critical evaluation of the original work. The experimental results showed that all evaluated machine learning models achieved good classification performance, confirming that engineered cybersecurity features are effective for distinguishing phishing websites from legitimate ones.

Lessons Learned

The project demonstrated the importance of data preprocessing, exploratory data analysis, feature engineering, and the selection of appropriate evaluation metrics. It also showed that evaluating models using multiple performance measures provides a more complete understanding of their strengths and weaknesses than relying only on accuracy.

Strengths

The main strengths of the proposed solution include its simple and well-organized workflow, meaningful engineered features, and comparison of several machine learning models. The provided code was easy to reproduce, and the experiments produced results that were highly consistent with the original project.

Weaknesses

The dataset contains only engineered numerical features and does not include webpage content or visual information. Therefore, the models may not detect all modern phishing techniques. In addition, the original project provides limited discussion regarding model explainability and long-term robustness against evolving cyber threats.

Future Improvements

Future work could improve the system by incorporating additional cybersecurity features, such as webpage content analysis, SSL certificate information, JavaScript behavior, domain registration history, and real-time threat intelligence. Evaluating deep learning approaches and testing the models on newer phishing datasets would also improve the practical applicability of the system.

8. Error Analysis

Examples of Model Failures

The confusion matrices showed that all evaluated models made both false positive and false negative predictions. For example, the MLP model incorrectly classified some legitimate websites as phishing (false positives) and failed to detect some phishing websites (false negatives). Although these errors were relatively few compared to the total number of predictions, they demonstrate that no model achieved perfect classification.

Patterns in the Errors

The results indicate that the models generally performed well on most samples but experienced difficulties with websites whose characteristics were similar to both phishing and legitimate websites. Such borderline cases are more challenging to classify because their engineered features do not strongly belong to a single class.

Cybersecurity Implications of the Errors

In cybersecurity, the consequences of classification errors are significant. A false positive incorrectly labels a legitimate website as phishing, which may inconvenience users and reduce trust in the detection system. A false negative is more serious because it allows a phishing website to be classified

as legitimate, potentially exposing users to credential theft, financial fraud, or identity theft.

The Trade-off Between False Positives and False Negatives

There is an important trade-off between reducing false positives and reducing false negatives. A highly conservative model may reduce false positives but miss more phishing websites, while a more aggressive model may detect more phishing attacks at the cost of generating additional false alarms.

In our experiments, the Multilayer Perceptron (MLP) achieved the best balance between these two types of errors. It produced high overall accuracy while maintaining strong recall, allowing it to detect a larger proportion of phishing websites without generating an excessive number of false positives. This balance makes the MLP model particularly suitable for phishing website detection.

9. Executive Summary

Phishing attacks continue to be one of the most significant cybersecurity threats, making accurate phishing website detection an important research problem. This project reproduced and critically evaluated an existing machine learning solution for phishing website detection using the dataset and implementation provided in the original GitHub repository.

The project began with data inspection and exploratory data analysis (EDA), including missing value analysis, duplicate analysis, feature distributions, class balance analysis, correlation analysis, and visualization of the dataset. The dataset contains engineered cybersecurity features extracted from website URLs and domain information, allowing machine learning algorithms to classify websites as phishing or legitimate.

Several machine learning models were trained and evaluated, including Decision Tree, Random Forest, Multilayer Perceptron (MLP), XGBoost, and

Support Vector Machine (SVM). Their performance was compared using multiple evaluation metrics, including Accuracy, Precision, Recall, F1-Score, ROC-AUC, Matthews Correlation Coefficient (MCC), and Confusion Matrix analysis.

The reproduction study successfully confirmed the main findings of the original project. Although the original notebook identified XGBoost as the best-performing model, our implementation achieved slightly higher performance with the Multilayer Perceptron (MLP). The difference between the two models was very small and did not affect the overall conclusions of the project.

Overall, the experiments demonstrated that engineered cybersecurity features combined with supervised machine learning provide an effective solution for phishing website detection. The project also highlighted the importance of using multiple evaluation metrics instead of relying only on accuracy when comparing classification models.

10. Summing It Up

The Problem Being Addressed

This project addresses the problem of phishing website detection. Phishing websites imitate legitimate websites in order to steal sensitive user information such as usernames, passwords, and financial data. Detecting phishing websites is an important cybersecurity challenge because phishing attacks continue to increase and can cause significant financial and personal damage.

The Selected Article / GitHub Project

The selected project is "**Phishing Website Detection by Machine Learning Techniques**", available on GitHub. The project investigates the use of

supervised machine learning algorithms to classify websites as either phishing or legitimate using engineered cybersecurity features.

The Dataset Used

The project uses a phishing website dataset containing 10,000 website samples. Each sample is represented by engineered numerical features extracted from URLs and domain characteristics. The target variable (Label) indicates whether the website is phishing or legitimate.

The Methodology Employed

The methodology consists of data preprocessing, exploratory data analysis (EDA), feature engineering, model training, and model evaluation. Five supervised machine learning models were trained and compared: Decision Tree, Random Forest, Multilayer Perceptron (MLP), XGBoost, and Support Vector Machine (SVM). The models were evaluated using Accuracy, Precision, Recall, F1-Score, ROC-AUC, Matthews Correlation Coefficient (MCC), and Confusion Matrix analysis.

The Main Findings of the Reproduction Study

The reproduction study successfully reproduced the original project and produced results that were highly consistent with the original implementation. Although the original project identified XGBoost as the best-performing model, our experiments showed that the Multilayer Perceptron (MLP) achieved slightly better performance. However, the difference between the two models was very small.

Were the Author's Claims Supported?

Yes. The experimental results support the author's main claim that machine learning techniques can effectively distinguish phishing websites from legitimate websites using engineered cybersecurity features. Although there was a small difference in the ranking of the best-performing models, the overall conclusions remained consistent with those presented in the original project.

The Most Important Insights Obtained from the Analysis

The analysis demonstrated that feature engineering plays a major role in phishing website detection. Carefully designed cybersecurity features provide sufficient information for machine learning algorithms to achieve high classification performance. In addition, evaluating models using multiple performance metrics provides a more reliable assessment than relying only on accuracy.

Recommendation

Yes. I recommend using this project as a baseline for similar phishing website detection problems. The project is well organized, reproducible, and easy to understand. However, future work should include newer datasets, additional cybersecurity features, explainable AI techniques, and continuous model updates to improve robustness against evolving phishing attacks.

Final Conclusion

Overall, this project successfully reproduced and critically evaluated the original phishing website detection study. The reproduction confirmed that the proposed methodology is appropriate for the problem and that the author's main claims are supported by experimental evidence. While there are

opportunities for improvement, the project provides an effective and reliable machine learning solution for phishing website detection.

References

[1] Shreya Gopal, "Phishing Website Detection by Machine Learning Techniques",

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